

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$U'_{[4] \times [N]} = - \begin{pmatrix} U'_{11} & U'_{12} & \cdots & U'_{1N} \\ U'_{21} & U'_{22} & \cdots & U'_{2N} \\ U'_{31} & U'_{32} & \cdots & U'_{3N} \\ U'_{41} & U'_{42} & \cdots & U'_{4N} \end{pmatrix},$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} a_{ij}^{\mathbf{4}} &= \frac{1}{g} \sum_{\gamma=1}^r r_{\gamma} \chi_{\gamma}^{R_M \oplus \mathbf{1}^2} \chi_{\gamma}^{(i)} \chi_{\gamma}^{(j)*} \\ a_{ij}^{\mathbf{6}} &= \frac{1}{g} \sum_{\gamma=1}^r r_{\gamma} \chi_{\gamma}^{R_M \oplus R_M \oplus \mathbf{1}^2} \chi_{\gamma}^{(i)} \chi_{\gamma}^{(j)*} \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \bar{W}_k(\sigma^+, \sigma^-) &= \left[\frac{1}{U_k(\sigma^+) + v_k} + \frac{1}{U_k(\sigma^-) + v_k} - \frac{1}{u_{k-1} + v_k} \right]^{-1} - v_k \\ U_{k+1}(z) &= \left[\frac{1}{U_k(z) + v_k} + \frac{1}{u_k + v_k} - \frac{1}{u_{k-1} + v_k} \right]^{-1} - v_k \\ u_{k+1} &= \left[\frac{2}{u_k + v_k} - \frac{1}{u_k + v_{k-1}} \right]^{-1} - v_k \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} & \sum_{i=0}^{l_1(x)-1} \{ \sum_{i'=0}^{l_2(\sigma_B^i(h(x)))-1} f(\sigma_C^{i'}(g(\sigma_B^i(h(x)))))) - \sum_{j'=0}^{k_2(\sigma_B^i(h(x)))-1} f(\sigma_C^{j'}(g(\sigma_B^i(h(x)))))) \} \\ &= \sum_{i'=0}^{l_2^{l_1(x)}(h(x))-1} f(\sigma_C^{i'}(gh(x))) - \sum_{j'=0}^{k_2^{l_1(x)}(h(x))-1} f(\sigma_C^{j'}(g(\sigma_B^{l_1(x)}(h(x)))))) \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \epsilon &= 1 & \delta \varepsilon_1^1\left(\frac{\pi}{5}\right) &= \delta \varepsilon_3^1\left(\frac{\pi}{5}\right), \\ & & \delta \varepsilon_1^1\left(\frac{\pi}{4}\right) &= \delta \varepsilon_2^1\left(\frac{\pi}{4}\right), \\ & & \delta \varepsilon_0^1\left(\frac{\pi}{4}\right) &= \delta \varepsilon_3^1\left(\frac{\pi}{4}\right), \end{aligned}$$